

ABTT Technical Standards

Mechanical Inspection Application Note

For Stage Electrics mechanical inspection work in places of entertainment

Field	Entry
Document status	Draft application note for internal review
Prepared for	Stage Electrics - mechanical inspection activity
Prepared date	26 May 2026
Primary source base	Publicly accessible ABTT, HSE and related trusted-source information listed in Appendix B
Critical limitation	The full ABTT Technical Standards text is a controlled/purchased resource. This note uses public descriptions and should be checked against the company's current licensed ABTT Yellow Book/online updates before being issued as a controlled procedure.

Use restriction: This document is deliberately limited to mechanical inspection work. It does not cover electrical inspection, emergency lighting, fire precautions, licensing sign-off, structural engineering design, general venue management, production operation, or repair work unless those activities are specifically included in a Stage Electrics contract and are undertaken by competent persons under the relevant procedure.

1. Executive summary

ABTT Technical Standards for Places of Entertainment, commonly referred to as the Yellow Book, are described by ABTT as a set of standards for buildings, equipment and management of premises primarily used for entertainment. ABTT states that the standards are intended mainly for indoor venues, with much guidance also applying to outdoor events, and are aimed at helping licensees, technical managers and enforcers understand places of entertainment requirements. [S1]

For Stage Electrics mechanical inspections, the practical application is not to reproduce the Yellow Book. The safer approach is to treat it as venue-context guidance and align the inspection process with statutory duties and recognised industry practice, particularly LOLER, PUWER, work-at-height expectations, manufacturer instructions, client records, examination schemes and applicable ABTT Codes of Practice. [S4-S10]

The minimum operational principle is that every inspection should answer one question clearly: can the specific mechanical item, in its observed condition and intended context, continue to be used safely until the next inspection or examination, subject to any limitations stated in the report?

2. Source basis and limitations

Trusted source	Relevant information identified	Application to Stage Electrics mechanical inspections
ABTT Technical Standards for Places of Entertainment	ABTT describes the publication as covering buildings, equipment and management of entertainment premises; current use should be checked against the latest edition and online updates. [S1-S3]	Use as the entertainment-venue context and benchmark. Do not treat public summaries as the complete standard.
ABTT Codes of Practice	ABTT states that its Codes of Practice provide more detailed guidance on technical topics than Technical Standards. Publicly listed examples include guard rails, temporary access equipment for work at height in theatres, Tallescopes, and theatre essentials. [S4]	When a Code of Practice applies to an inspected asset or access method, use it as a specific industry reference alongside law, manufacturer instructions and Stage Electrics procedure.

HSE theatre guidance	HSE says health and safety law applies to theatres; most theatre injuries are related to work at height and manual handling; HSE points to ABTT guidance for temporary access equipment in theatres. [S5]	Plan inspection access and client handover for theatre-specific hazards, especially work at height, manual handling, poor lighting, restricted access and venue operation.
LOLER and HSE lifting-equipment guidance	Lifting equipment must be fit for purpose, suitable, marked with SWL, planned/supervised by competent people, and in many cases subject to statutory thorough examination and reporting. [S6-S7]	Primary framework for hoists, winches, lifting points, counterweight/flying systems, lifting accessories and similar equipment used at work to lift/lower loads or persons.
PUWER and HSE work-equipment inspection guidance	Work equipment must be suitable, safe, maintained, inspected where needed, and used by trained persons; inspection scope/frequency should be risk based. [S8-S9]	Applies to mechanical work equipment even where LOLER does not apply, including machinery, access equipment, stage machinery, guarding and safety-related mechanisms.
Work at Height Regulations / HSE guidance	Work at height must be properly planned, supervised and carried out by competent people using suitable equipment; risks must first be assessed. [S10-S11]	Applies to the inspection method itself and to any access equipment inspected as part of the job.

Known gap: No Stage Electrics internal procedure, asset taxonomy, competency matrix or report template was available to the author of this note. Before adoption, this document should be cross-checked against Stage Electrics controlled procedures, insurance requirements, accreditations, client contract terms and the current licensed ABTT Technical Standards text.

3. Scope

3.1 In scope

- Mechanical inspection and/or thorough examination of entertainment workplace equipment used to lift, lower, suspend, support, move, restrain, guide or protect loads, performers, technicians or audience members.
- Examples: counterweight or hemp flying systems, powered flying equipment, hoists, chain blocks, winches, lifting accessories, suspension points, stage engineering machinery, traps, turntables, orchestra or platform lifts, guard rails, access equipment and associated mechanical safety devices where included in the contracted inspection scope.
- Inspection planning, evidence review, competent-person judgement, defect categorisation, reporting, immediate-danger escalation and retention of records.

3.2 Out of scope unless separately contracted

- Electrical inspection, PAT, fixed wiring, emergency lighting, control-system validation, software validation or electrical safety testing.
- Structural calculations, design certification, load-rating creation, asbestos surveys, fire strategy, licensing approval, crowd-management assessment, general risk assessment for the client's venue operation, or independent auditing of the client's safety-management system.
- Maintenance or repair work by the same person who is providing impartial thorough examination, unless independence and conflict-of-interest controls have been agreed and recorded.
- Operation of lifting equipment for production purposes or lift planning, unless this is expressly included and assigned to competent persons.

4. Key definitions for inspectors

Term	Working definition for this document
Mechanical inspection	A planned visual, functional and evidence-based assessment of mechanical equipment, including any measurement, test, dismantling or further examination justified by risk, manufacturer information, industry practice or competent-person judgement.
Thorough examination	A systematic and detailed examination of lifting equipment and safety-critical parts at specified intervals by a competent person, followed by a written report containing the required information. [S7]

Competent person	A person with appropriate practical and theoretical knowledge and experience of the equipment to detect defects or weaknesses and assess their importance for safety and continued use. [S7]
Dutyholder	The client, employer, owner, operator, hirer, controller of premises or other person with legal responsibility for the equipment and work activity.
Lifting accessory	Equipment used to connect a load to lifting equipment, such as a sling, shackle, clutch, beam clamp or similar item, depending on context and design.
Safe working load / working load limit	The maximum load the equipment or accessory is marked and intended to lift safely in a specified configuration. LOLER requires clear marking of SWL for lifting equipment and accessories. [S6]

5. How to apply ABTT Technical Standards without overextending them

5.1 Apply the ABTT material as venue-context guidance

- Use the ABTT Technical Standards as a places-of-entertainment context document: theatres and entertainment venues have unusual hazards caused by performance conditions, work at height, moving scenery, flying systems, temporary arrangements, darkness, noise, public occupancy and restricted backstage access. [S1, S5]
- Use ABTT Codes of Practice where the public listing or the company's licensed resources confirm a relevant code applies to a specific technical topic. ABTT states these codes are typically more detailed on specific subjects than the Technical Standards. [S4]
- Treat the ABTT publication as one part of the evidence base, not as a replacement for LOLER, PUWER, Work at Height Regulations, manufacturer instructions, recognised standards, insurer requirements or the client contract. [S6-S11]

5.2 Do not apply unrelated Yellow Book content

- Do not import requirements for fire, emergency lighting, electrical installation, licensing, seating, public circulation, noise, safeguarding, building fabric or venue operation into a mechanical inspection report unless the item directly affects the inspected mechanical equipment or the inspector's safe access.
- Do not imply that Stage Electrics has completed a full ABTT Technical Standards compliance audit unless that broader service was contracted and performed by competent persons against the full current ABTT text.
- Do not copy ABTT text into Stage Electrics documents unless the company has permission and the copied content is controlled for updates.

6. Mechanical inspection workflow

6.1 Pre-visit review

1. Confirm contracted scope, site location, asset list, requested inspection type and whether the work is a LOLER thorough examination, PUWER inspection, work-at-height equipment inspection, condition survey, or another defined service.
2. Request previous reports, defect close-out records, manufacturer manuals, declarations of conformity/incorporation, rated load information, examination schemes, load test evidence where relevant, maintenance records, modification records and access arrangements.
3. Check whether any ABTT Code of Practice, manufacturer instruction, client standard or recognised technical standard applies to the asset type.
4. Plan safe access. Apply the HSE work-at-height hierarchy and ensure the person undertaking the work has sufficient skills, knowledge and experience for both the equipment and the access method. [S10-S11]
5. Identify limitations likely to affect the inspection: inaccessible parts, poor lighting, missing documents, inability to operate equipment, occupied venue, asbestos restrictions, time limitations or need for specialist testing.

6.2 On-site opening meeting

6. Confirm the dutyholder/client representative, site induction, permit requirements, emergency arrangements, isolation points, exclusions, rescue arrangements and whether equipment is in service or already isolated.
7. Confirm the asset list and any changes since quotation. Record additions, omissions and equipment not found.
8. Agree how immediately dangerous defects will be escalated during the visit and who has authority to quarantine equipment.

6.3 Inspection/examination activity

9. Verify identity, markings, SWL/WLL, serial numbers, locations and configuration against the asset list and previous records.
10. Inspect only within competence. Where a defect, asset type or configuration is outside competence, stop and escalate for specialist advice.
11. Use the least intrusive method sufficient for safety judgement, but include functional checks, measurement, dismantling, NDT or testing when justified by risk, manufacturer instructions, examination scheme or competent-person judgement. HSE notes that routine load testing is not normally required for most lifting equipment and can cause damage if unnecessary. [S7]
12. Record limitations clearly. If a safety-critical part cannot be examined, do not imply that it has passed.
13. Where immediate or potentially dangerous defects are identified, notify the dutyholder immediately and record the notification. For LOLER, HSE states that serious defects must be reported verbally, followed by a written report, and that relevant enforcing-authority notification is required in specified cases. [S7]

6.4 Reporting and handover

14. Issue a written report that is clear enough for the dutyholder to understand what was inspected, what was not inspected, what defects were found, what action is needed, and whether the equipment can continue to be used.
15. For LOLER thorough examinations, use a report template that captures the Schedule 1 information required by LOLER. HSE confirms there is no longer a defined form, provided all required Schedule 1 items are included. [S7]
16. State the next inspection or thorough-examination due date, or state that continued use is not permitted until specified action is completed.
17. Retain records securely and protect them from unauthorised alteration. HSE says records may be electronic but must be available as written copies when required. [S7-S9]

7. Inspection and thorough-examination timing

Trigger / equipment type	Baseline expectation	Source basis
Before first use of lifting equipment	Thorough examination before first use unless a valid Declaration of Conformity less than 12 months old applies and safety does not depend on site assembly/installation.	HSE LOLER guidance [S7]
After assembly or installation at a new location	Inspect/thoroughly examine before use where safety depends on installation or reassembly.	HSE LOLER and PUWER guidance [S7-S9]
Lifting equipment and accessories used to lift people	Thorough examination every 6 months unless a competent-person examination scheme sets a different justified interval.	HSE LOLER guidance [S7]
Lifting accessories	Thorough examination every 6 months unless a competent-person examination scheme sets a different justified interval.	HSE LOLER guidance [S7]
Other lifting equipment	Thorough examination every 12 months unless a competent-person examination scheme sets a different justified interval.	HSE LOLER guidance [S7]
PUWER work equipment exposed to deterioration	Inspect at suitable intervals determined through risk assessment, manufacturer recommendations, industry advice and experience.	HSE PUWER inspection guidance [S9]

After exceptional circumstances	Inspect or thoroughly examine after damage, failure, long out-of-use periods, major modification, repair/replacement of critical parts or substantial change in use.	HSE LOLER and POWER guidance [S7-S9]
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Stage Electrics rule of thumb: never extend an interval beyond the statutory default, manufacturer requirement, client requirement or previous competent-person requirement unless a competent person has produced and approved a written examination scheme or documented risk-based justification.

8. Competence, impartiality and handover controls

- Match the inspector to the asset class. Competence for general access equipment does not automatically prove competence for powered flying systems, performer flying, stage lifts, counterweight systems or complex automation-related mechanical systems.
- A person undertaking a thorough examination should be sufficiently independent and impartial. HSE notes that this should not be the same person who undertakes routine maintenance of the equipment where that would involve assessing their own maintenance work. [S7]
- Keep a competence matrix covering asset class, inspection type, limitations, training, experience, mentoring/sign-off and refresher requirements.
- When a defect is found, the inspector may explain the risk and reporting category, but the dutyholder remains responsible for preventing use and managing repairs unless Stage Electrics is separately contracted to do so.

9. Defect categories for Stage Electrics reports

Category	Meaning	Required report language / action
A - Immediate danger	A defect or condition that is, or could imminently become, a danger to people.	State that the item must not be used until remedied and rechecked as appropriate. Notify the dutyholder immediately. For LOLER serious defects, follow enforcing-authority reporting requirements.
B - Use restricted / action by date	A defect or deterioration that does not require immediate cessation in the competent person's judgement but must be remedied by a stated date or before further specified use.	State the limitation, latest remedy date and consequence if not completed.
C - Observation / monitor	A non-dangerous observation, early deterioration, documentation gap or good-practice recommendation.	Record plainly without implying statutory failure. Consider shortening interval if trend indicates deterioration.
D - Not examined / limitation	The item or part could not be inspected, tested or operated sufficiently to support a safety judgement.	State the limitation, reason and action needed to complete the inspection. Avoid pass/fail wording for unexamined parts.

10. Asset-focused inspection matrix

Asset group	Mechanical inspection focus	Typical evidence / notes
All mechanical assets	Identity, SWL/WLL, location, configuration, visible damage, corrosion, deformation, security of fixings, guarding, emergency access, safe isolation, previous defects, maintenance history and user-reported issues.	Manufacturer instructions, previous reports, drawings, load data, modification records, client asset register.
Counterweight, hemp and manual flying systems	Bars, cradles, arbors, guides, locks, brakes, head/loft blocks, sheaves, bearings, rope/wire condition, terminations, balance/trim issues, travel path, guards, signage and operating restrictions.	Confirm what can be seen from safe access. Record if any high-level components cannot be examined.
Powered hoists, winches and stage-lifting machinery	Suspension/support, body, hooks, chains/wire ropes, drums, brakes, gearboxes, limit devices where mechanically inspectable, guards, emergency lowering/holding arrangements, unusual noise, smooth travel and evidence of overload/misuse.	Mechanical inspection may need interface with electrical/control specialists. Do not certify control-system integrity unless in scope and competent.
Lifting accessories	Identification, SWL/WLL, traceability, compatibility, distortion, cracks, corrosion, wear, broken wires, kinks, heat or chemical damage, seized parts, missing latches, damaged threads, illegible markings and discard criteria.	Use manufacturer discard criteria and applicable recognised standards. Lifting accessories normally require 6-month thorough examination under LOLER unless scheme states otherwise. [S7]
Fixed suspension points, beams, clamps and supports	Identification, rated capacity, configuration, anchorage, movement, deformation, corrosion, missing fasteners, compatibility, secondary suspension and evidence of unauthorised modification.	Structural adequacy is outside scope unless engineering design/certification is included. Report lack of evidence as a limitation.
Guard rails, barriers and edge protection	Height/coverage, rigidity, fixings, gates, toe boards where applicable, gaps, corrosion, impact damage and suitability for foreseeable backstage use.	ABTT publicly lists a Design Guide: Guard Rails Code of Practice. Use current licensed copy if applicable. [S4]
Tallescopes, towers, ladders and temporary access equipment	General condition, outriggers/stabilisers, castors, brakes, platforms, guardrails, locking devices, labels, records and suitability for inspection access.	ABTT publicly lists Codes of Practice for temporary access equipment and Tallescopes in theatres. HSE theatre guidance links to ABTT temporary access guidance. [S4-S5]
Traps, turntables, revolves, wagons and guided scenic mechanisms	Guidance tracks, bearings, rollers, pivots, locks, stops, guards, pinch/crush points, condition of decks, emergency release or hold points, manual handling risks and maintenance condition.	Often PUWER-led rather than LOLER-led unless lifting/lowering is involved. Confirm which regulations apply to the actual equipment and use.

11. Inspection method notes

11.1 Evidence hierarchy

- Current law and HSE guidance.
- Current ABTT Technical Standards and any relevant ABTT Code of Practice, using Stage Electrics licensed/current copies where necessary.
- Manufacturer instructions, declarations, rated-capacity data and maintenance instructions.
- Examination scheme drawn up by a competent person.
- Recognised British/European/industry standards, insurer requirements and client contractual requirements.
- Competent-person judgement based on inspection evidence and inspection history.

11.2 What to avoid

- Do not pass an item solely because it was previously passed.

- Do not fail an item solely because documentation is incomplete; instead, decide whether the missing evidence prevents a safe-use judgement and record the limitation or required action.
- Do not carry out proof/load testing routinely unless the competent person determines it is required. HSE warns that some overload tests can damage lifting equipment. [S7]
- Do not rely on a tick-box checklist where unusual venue conditions or asset deterioration require deeper competent-person judgement.

12. Recommended Stage Electrics report content

Report section	Minimum content
Client and site details	Client/dutyholder name, site address, venue area, client representative, inspection date(s) and job reference.
Scope and standard basis	State whether the work is a LOLER thorough examination, PUWER inspection, condition survey or other service. List source basis used and exclusions.
Asset identification	Unique ID, description, manufacturer, model, serial number, SWL/WLL, location, configuration, use category and previous report reference.
Documents reviewed	Manufacturer instructions, DoC, previous reports, maintenance records, examination scheme, modifications and any documents not available.
Inspection method	Visual, functional, measurement, dismantling, NDT, load test or other method used. State operating limitations and access limitations.
Findings	Defects, observations, photographs if used, risk category, required action, responsible party and timescale.
Continued-use judgement	Clear statement: safe to continue until next due date; restricted use only; not to be used until remedied; or no judgement due limitation.
Next due date	Date of next thorough examination or inspection, and basis for the interval.
Sign-off	Inspector/competent person name, role, company, signature/approval, date issued and version control.

13. Implementation checklist before this note becomes controlled

- Cross-check against Stage Electrics' current licensed ABTT Technical Standards and online updates.
- Cross-check against Stage Electrics' existing LOLER/PUWER reporting templates and ensure LOLER Schedule 1 information is captured.
- Map each asset class to an approved competence level and authorised inspector list.
- Confirm report wording for immediate-danger and enforcing-authority notification with Stage Electrics' compliance lead/insurer.
- Confirm document owner, review period and change-control process.
- Link this note to controlled forms: pre-visit request, inspection checklist, defect report, quarantine/return-to-service form and client handover record.

Appendix A - Quick inspection prompt sheet

Prompt	Inspector check
Identity	Is the asset uniquely identifiable and does it match the asset register?
Rating	Is SWL/WLL clear, legible and valid for the actual configuration?
Use category	Is it lifting persons, lifting accessories, other lifting equipment, PUWER work equipment, access equipment or another category?
Documentation	Are previous reports, maintenance records, DoC/manuals and examination scheme available?
Condition	Any deformation, corrosion, cracking, wear, broken wires, damaged chains, missing fasteners, seized components or abnormal movement?
Function	Does it operate smoothly and stop/hold/control loads as expected where operation is safe and in scope?
Safety devices	Are guards, stops, brakes, locking devices, warning markings and interlocks present and apparently effective within the mechanical scope?
Access limitations	Which parts could not be seen, reached, operated or verified?
Defect action	Is the dutyholder informed immediately of dangerous defects? Has the report stated not to use/restrictions/repair-by date?
Next due date	Is the next interval justified by LOLER/PUWER, manufacturer recommendations, scheme, history and risk?

Appendix B - Sources consulted

- **S1:** Association of British Theatre Technicians (ABTT). Technical Standards for Places of Entertainment - product page, hard copy with online access. <https://www.abtt.org.uk/product/technical-standards-for-places-of-entertainment-hard-copy-with-access-to-e-book/>
- **S2:** ABTT. Technical Standards for Places of Entertainment - resources page / online viewing information. <https://www.abtt.org.uk/resources/technical-standards-for-places-of-entertainment/>
- **S3:** The Construction Information Service / Accuris. Technical standards for places of entertainment (New edition July 2022), status Current. <https://cis.ihs.com/CIS/document/339129>
- **S4:** ABTT. Shop - Codes of Practice category page. <https://www.abtt.org.uk/product-category/codes-of-practice/>
- **S5:** Health and Safety Executive (HSE). Theatre guidance. <https://www.hse.gov.uk/entertainment/theatre-tv/theatre.htm>
- **S6:** HSE. Lifting Operations and Lifting Equipment Regulations 1998 (LOLER) - overview. <https://www.hse.gov.uk/work-equipment-machinery/loler-overview.htm>
- **S7:** HSE. Thorough examinations and inspections of lifting equipment. <https://www.hse.gov.uk/work-equipment-machinery/thorough-examinations-lifting-equipment.htm>
- **S8:** HSE. Provision and Use of Work Equipment Regulations 1998 (PUWER) - overview. <https://www.hse.gov.uk/work-equipment-machinery/puwer-overview.htm>
- **S9:** HSE. Inspection of work equipment. <https://www.hse.gov.uk/work-equipment-machinery/inspection.htm>
- **S10:** HSE. Work at height - The law. <https://www.hse.gov.uk/work-at-height/the-law.htm>
- **S11:** HSE. Working at height: A brief guide INDG401. <https://www.hse.gov.uk/pubns/indg401.pdf>

Appendix C - Adoption note

This document is suitable as a starting point for a Stage Electrics controlled procedure or toolbox guidance note. It should not be issued externally as evidence of ABTT compliance until checked against the current licensed ABTT Technical Standards and approved internally by the relevant Stage Electrics technical/compliance owner.